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Lecture #5 Nearly free electron model

Problem 5.1 (Compare Kronig-Penney model with the nearly free electron model). The Kronig-Penney model allows one to consider both the nearly-free-electron and tight-binding limits. Consider the (repulsive) Dirac-Kronig-Penney model. It gives the following implicit equation for the energy bands

$$\cos(ka) = \cos(qa) + u \frac{\sin(qa)}{qa}$$

where $q = \sqrt{2mE}/\hbar$ and $u = \frac{mU_0a}{\hbar^2} > 0$. Assuming that $u \ll q$, find the solution of above equation near the Bragg points $k \approx \pm \frac{\pi}{a}, \pm \frac{2\pi}{a}, \dots$. Compare your result with that of the nearly-free electron model. (Reminder: the $\approx$ sign means that k is close yet not equal to the wavenumber of the Bragg point.)

Solution. Near the Bragg points, assume

$$ka = n\pi + \xi a, \quad \text{and} \quad qa = q_n a + \delta$$

with $|\xi|, |\delta| \ll 1$, $q_n = n\pi/a$. Keep the 2nd order of ξ and δ , the implicit equation can be written as

$$(-1)^n \left[1 - \frac{1}{2} (n\pi)^2 \right] = (-1)^n \left(1 - \frac{1}{2} \delta^2 \right) + u \frac{(-1)^n \delta}{n\pi}$$

Then, we can have the solution of δ

$$\delta = \frac{u}{n\pi} \pm \sqrt{\left(\frac{u}{n\pi} \right)^2 + (\xi a)^2}$$

Since $q = \sqrt{2mE}/\hbar$, then we have

$$E = \frac{\hbar^2 (q_n + \delta/a)^2}{2m} \approx E_n^0 + \frac{\hbar^2 q_n \delta/a}{m}$$

where $E_n^0 = \frac{\hbar^2 q_n^2}{2m}$. Substitute $q_n = n\pi/a$, and the solution of δ into the expression of E

$$E = E_n^0 + \frac{n\pi \hbar^2 \delta}{ma} = E_n^0 + \frac{n\pi \hbar^2}{ma} \left[\frac{u}{n\pi} \pm \sqrt{\left(\frac{u}{n\pi} \right)^2 + (\xi a)^2} \right]$$

Substitute $u = mU_0 a \hbar^2$ into the expression above

$$E = E_n^0 + \frac{U_0}{a} \pm \sqrt{\left(\frac{U_0}{a} \right)^2 + \left(\frac{n\pi \hbar^2 \xi}{ma} \right)^2}$$

with the band gap

$$\Delta E = E_+ - E_- = 2 \sqrt{\left(\frac{U_0}{a} \right)^2 + \left(\frac{n\pi \hbar^2 \xi}{ma} \right)^2}$$

When $\xi a \rightarrow 0$, the Energy band and the band gap become

$$E = E_n^0 + \frac{U_0}{a} \pm \frac{U_0}{a}, \quad \Delta E = \frac{2U_0}{a}$$

The results perfectly satisfy that of the N.F.E. model.

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Diagonalizing the matrix gives the eigenenergies

$$E_{\pm} = E_0 \pm \sqrt{(\alpha q_x)^2 + \left(\frac{U_0}{2}\right)^2} = E_0 \pm \sqrt{\left(\frac{\hbar^2 a^2}{ma} q_x\right)^2 + \left(\frac{U_0}{2}\right)^2} + \frac{\hbar^2 q_x^2}{2m}.$$

(b). The corners of the 1st BZ are points like $\vec{k}_0 = (\frac{\pi}{a}, \frac{\pi}{a})$, i.e., the X -points. The four degenerate plane waves at this point are

$$\vec{k}_1 = \vec{k}_0 = \left(+\frac{\pi}{a}, +\frac{\pi}{a}\right)$$

$$\vec{k}_2 = \vec{k}_0 - \vec{G}_1 = \left(-\frac{\pi}{a}, +\frac{\pi}{a}\right)$$

$$\vec{k}_3 = \vec{k}_0 - \vec{G}_2 = \left(+\frac{\pi}{a}, -\frac{\pi}{a}\right)$$

$$\vec{k}_4 = \vec{k}_0 - \vec{G}_1 - \vec{G}_2 = \left(-\frac{\pi}{a}, -\frac{\pi}{a}\right)$$

All have the same free-electron energy $E_0 = \frac{\hbar^2}{2m} \left[\left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{a}\right)^2 \right] = \frac{\hbar^2 \pi^2}{ma^2}$.

In the basis $\{|\vec{k}_1\rangle, |\vec{k}_2\rangle, |\vec{k}_3\rangle, |\vec{k}_4\rangle\}$, states are coupled if they differ by $\pm \vec{G}_1$ or $\pm \vec{G}_2$. The non-zero off-diagonal matrix elements are

$$\langle \vec{k}_1 | U | \vec{k}_2 \rangle = \langle \vec{k}_1 | U | \vec{k}_3 \rangle = \langle \vec{k}_2 | U | \vec{k}_4 \rangle = \langle \vec{k}_3 | U | \vec{k}_4 \rangle = U_0/2.$$

The Hamiltonian is

$$H_0 = E_0 I + \frac{U_0}{2} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

Diagonalizing this matrix yields eigenvalues

$$E = E_0 + \frac{U_0}{2} \times \{2, -2, 0, 0\} = \{E_0 + U_0, E_0 - U_0, E_0, E_0\}.$$

Thus, at $\vec{q} \neq 0$, the fourfold degeneracy splits into three distinct levels: ~~two~~ non-degenerate bands at $E_0 + U_0$ and $E_0 - U_0$, and a two-fold degenerate band at E_0 .

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Problem 5.2 (Nearly free electron model). Consider 2D electrons subject to a weak periodic potential

$$U(x, y) = U_0 \left(\cos \frac{2\pi x}{a} + \cos \frac{2\pi y}{a} \right).$$

- (a) Find the energy bands near the edges of the first Brillouin zone. (Hint: the edges of the Brillouin zone are the Bragg planes where the eigenstates are doubly degenerate.) Plot the bands and isoenergetic surfaces.
- (b) Repeat the analysis of part (a) for regions near the corners of the first Brillouin zone, where there are four degenerate eigenstates.

Solution.

(a) In reciprocal space, the reciprocal lattice vectors are

$$\vec{G}_1 = \left(\frac{2\pi}{a}, 0 \right), \quad \vec{G}_2 = \left(0, \frac{2\pi}{a} \right).$$

Then, the potential can be expressed as

$$U(x, y) = \frac{U_0}{2} \left[e^{i\vec{G}_1 \cdot \vec{r}} + e^{-i\vec{G}_1 \cdot \vec{r}} + e^{i\vec{G}_2 \cdot \vec{r}} + e^{-i\vec{G}_2 \cdot \vec{r}} \right]$$

At the zone edge (i.e., $k_x = \pi/a$), consider a wave vector

$$\vec{k} = \left(\frac{\pi}{a} + q_x, k_y \right)$$

with $|q_x| \ll \pi/a$. The two nearly degenerate plane waves are $|\vec{k}\rangle$ and $|\vec{k} - \vec{G}_1\rangle$. Their free electron energies are

$$E^{(0)}(\vec{k}) = \frac{\hbar^2}{2m} \left[\left(\frac{\pi}{a} + q_x \right)^2 + k_y^2 \right], \quad E^{(0)}(\vec{k} - \vec{G}_1) = \frac{\hbar^2}{2m} \left[\left(-\frac{\pi}{a} + q_x \right)^2 + k_y^2 \right].$$

Expanding to the first order in q_x

$$E^{(0)} \approx E_0 \pm \alpha q_x, \quad \text{where } E_0 = \frac{\hbar^2}{2m} \left[\left(\frac{\pi}{a} \right)^2 + k_y^2 \right], \quad \alpha = \frac{\hbar^2 \pi}{ma}.$$

The perturbation couples these states with matrix elements $\langle \vec{k} | U | \vec{k} - \vec{G}_1 \rangle = U_0/2$. The Hamiltonian in the subspace $\{|\vec{k}\rangle, |\vec{k} - \vec{G}_1\rangle\}$ is

$$\mathcal{H} = \begin{pmatrix} E_0 + \alpha q_x & U_0/2 \\ U_0/2 & E_0 - \alpha q_x \end{pmatrix}$$

